

Front-Loading Cost Estimation in Early-Stage BIM Design

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ABSTRACT

Artificial intelligence (AI) models are accelerating design exploration, yet early-stage feedback regarding project targets remains unavailable until major decisions are made even with building information modeling (BIM) adoption. One reason is the distributional under-representation of early-stage deployment state input features in available training data. Current estimators trained on finalized project data are prone to extrapolation violations when deployed in early design stage. This paper proposes a t-dependent origin anchoring transformation, an inference-time shape-constraining technique, that controls extrapolation in unseen input distributions while preserving estimation performance on seen distributions. The transformation regulates the exploration-exploitation trade-off using parameters defined at the input-feature and concatenated-configuration levels. Validation results demonstrate reduction in monotonicity violations by 84% and complete elimination of zero-state errors, while preserving or improving cost estimation accuracy across 94 BIM authoring simulations. These results suggest that finalized-project data can be leveraged to train estimators that remain reliable for early-stage AI-augmented BIM decision-making.

INTRODUCTION

The MacLeamy curve (CURT, 2004) describes an inverse relationship in construction projects, where the cost of design changes increases sharply as a project advances from pre-design to construction, while the ability to influence overall outcomes decreases correspondingly (Figure 1). Building information modeling (BIM) workflows address this challenge by shifting the authoring of three-dimensional geometry and semantic attributes to earlier stages (Borrmann et al., 2018), thereby providing a shared data source for early-phase informed-decision making.

Recent advances in artificial intelligence (AI) have further accelerated this shift by supporting various forms of design automation, including design option generation, layout refinement, and detailing (Jang et al., 2025; Darko et al., 2020). As a result, designers can explore a larger design space and iterate alternatives more rapidly during the project's early stages. However, the expansion of design alternatives in early design stage also intensifies the need for decision-making criteria grounded in expected project targets, most notably construction cost.

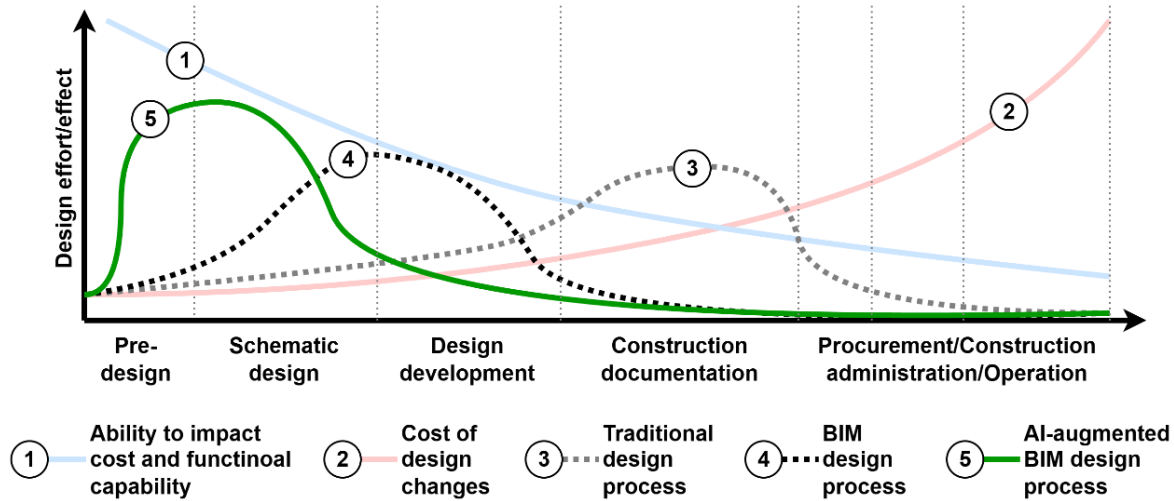


Figure 1. McLeamy curve projected on traditional 2D, BIM, and AI-augmented-BIM-based workflows, updated from CURT (2024).

In this context, earlier access to cost feedback has the potential to substantially influence and improve BIM-based early design stage decision-making (Tommelein and Ballard, 2016). In practice, however, cost estimation is usually performed only at specific project milestones using consolidated model states, while AI-based estimators are mostly trained on finalized or nearly complete projects. Early-stage BIM states are underrepresented in both estimation workflows and training distributions. Against this background, this study defines information front-loading as the integration of performance evaluation directly into the early design authoring process (Figure 1).

However, when such estimators are applied in early design stages, they usually need to predict beyond their training data range (Ribeiro et al., 2016). Such extrapolation often leads to unplausible or logically inconsistent results where design decisions have the greatest leverage. To address this limitation, this paper proposes a shape-constrained estimation method that explicitly regulates extrapolation behavior while preserving predictions within the training data distribution. The proposed t-dependent transformation enforces strong domain-consistent constraints in near-empty states and progressively relaxes them as inputs approach the seen distribution. The method is validated using 94 simulated BIM authoring mockup logs, demonstrating that extrapolation violations can be reduced without degrading estimation accuracy.

The remainder of this paper is organized as follows. Section 2 reviews related work and background on extrapolation issues in project performance estimation. Section 3 introduces the proposed t-dependent origin anchoring transformation, Section 4 describes the experimental design and evaluation metrics, Section 5 presents the results, and Section 6 concludes the paper.

BACKGROUND

Early-stage cost estimation is critical in design-to-target decision-making (Tommelein and Ballard, 2016), yet remains inherently uncertain due to sparse and incomplete information (Alqahtani and Whyte, 2016). In practice, early estimates often rely on coarse project-scale indicators such as gross floor area or functional unit counts. While these methods provide rapid overview, their uncertainty is substantial, commonly ranging between -30% and $+50\%$ (Bredehoeft et al., 2019), which limits their usefulness precisely when design flexibility and decision leverage are highest.

To improve estimation accuracy, extensive research and practice have focused on BIM-based cost estimation methods that exploit detailed geometric and semantic information: traditional approaches typically link bills of quantities extracted from BIM models with unit price or productivity databases, relying on predefined element classification schemes and curated cost standards (Ma et al., 2013). These methods perform well once design models are sufficiently detailed, but they are difficult to apply during early or evolving design stages, where classifications and activity definitions remain incomplete or uncertain. More recently, feature-based and AI-based estimation methods have been proposed to reduce dependence on fully specified BIM metadata (Elmousalami, 2020). These approaches predict aggregated project performance directly from compact feature vectors derived from BIM quantities, such as element counts, areas, and volumes (Park and Yun, 2023). While such methods improve flexibility and scalability, they are typically trained on finalized or near-complete project data and applied at discrete design milestones rather than during continuous BIM authoring.

This training–deployment mismatch becomes critical in early design stages. When AI-based estimators trained on late-stage project data are queried on partially authored BIM models characterized by sparse, low-count input features, they operate far outside the training distribution, as illustrated based on our simulation in Figure 2 (from the results of validation baseline).

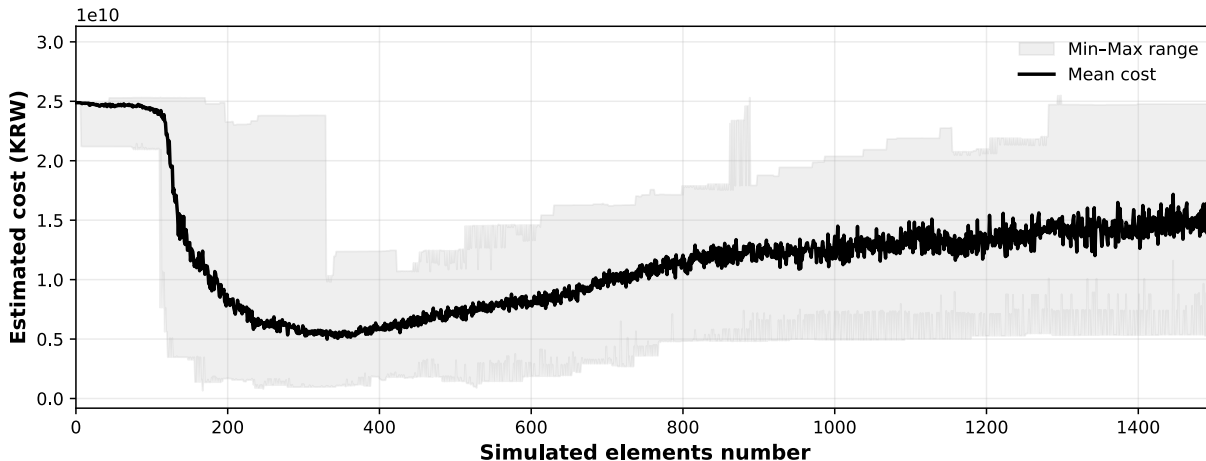


Figure 2. Extrapolation error observed when using AI estimators on an unseen input distribution.

These early design states occupy a weakly constrained region of the feature space. As a result, estimators are forced to extrapolate and often produce logically inconsistent predictions, such as non-zero estimates for near-empty states or violations of positive monotonicity. These extrapolation failures are not primarily caused by insufficient model capacity, but by the absence of explicit mechanisms that regulate estimator behavior outside the training distribution. Existing BIM-based estimation approaches have focused on predictive accuracy within the observed data distribution. Yet early-stage deployment conditions remain largely underexplored, although reliable feedback has the greatest potential to influence project outcomes at this stage.

In the broader machine learning literature, related challenges have been addressed through several complementary approaches. Out-of-distribution (OOD) detection methods aim to identify inputs that deviate from the training distribution, enabling selective abstention or uncertainty flagging (Hendrycks and Gimpel, 2017). Calibration techniques, such as temperature scaling and Platt scaling, adjust model confidence post-hoc to better reflect true predictive uncertainty (Guo et

al., 2017). Shape-constrained regression, including isotonic regression and monotonic neural networks, enforces global monotonicity through architectural or optimization constraints (Wehenkel and Louppe, 2019). However, these approaches either require retraining, impose constraints globally rather than adaptively, or focus on uncertainty quantification rather than output correction. The proposed method differs by introducing localized, input-dependent shape control at inference time without modifying the underlying estimator.

In BIM authoring, these prerequisites are not satisfied. Early design states yield no observable cost estimation, alternative designs cannot be repeatedly evaluated, and no feedback signal is available to correct extrapolation errors. Therefore, the key challenge is not to improve exploration during training, but to control model behavior during inference in weakly supported early design states. From this perspective, early-stage cost estimation requires a balance between exploiting learned relationships in seen distribution and enforcing domain-consistent behavior in unseen distribution. This motivates estimation methods that explicitly control extrapolation through parameterized shape constraints.

T-DEPENDENT ORIGIN ANCHORING TRANSFORMATION

We introduce a t -dependent origin anchoring transformation applied at inference time to regulate extrapolation behavior in early-stage BIM authoring, while preserving in-distribution predictions. The transformation is driven by a scalar progress index $t(x) \in [0, 1]$ computed from the input feature vector x , which quantifies how strongly the current state is supported by the training distribution. Values of $t(x)$ close to one indicate an exploitation regime where learned relationships are trusted, whereas values approaching zero correspond to an exploration-dominated regime in which predictions are constrained toward domain-consistent behavior. Let Q denote a set of quantity-type input features expected to evolve monotonically with project performance, such as element counts, areas, or volumes. For each feature, a reference scale τ is defined as a lower quantile of its training distribution, separating sparse early inputs from sufficiently supported values. Feature-wise normalized scores are aggregated and further normalized by r_{seen} . Intuitively, it defines the aggregate feature volume threshold below which inputs are treated as effectively unseen. Accordingly, larger the values expand this unseen region, imposing stronger shape constraints on a broader range of inputs. The resulting progress index is defined as:

$$t(x) = \text{clip}\left(\frac{1}{|Q|} \sum \min\left(1, \max\left(0, \frac{x}{\tau}\right)\right) / r_{seen}, 0, 1\right)$$

The parameters τ and r_{seen} jointly control how aggressively extrapolation is constrained. While τ governs the feature-wise transition from sparse to supported inputs, r_{seen} determines the extent of the aggregated region treated as unseen. At inference time, the progress index is used to modulate the raw estimator output. Predictions are damped toward zero for weakly supported states and progressively converge to the unconstrained estimator as inputs enter the seen distribution. This simple multiplicative modulation encodes the exploration-exploitation trade-off directly into the estimation process, enforcing domain-consistent behavior where training support is weak while preserving learned relationships where sufficient data support exists. From an information front-loading perspective, the proposed transformation provides a continuous and interpretable mechanism for regulating estimator behavior during early-stage design. In the following section, we empirically examine how different choices of τ and r_{seen} affect extrapolation stability and estimation accuracy.

VALIDATION

Validation examines whether the proposed t -dependent origin anchoring transformation suppresses extrapolation-induced shape violations in early-stage BIM authoring while preserving estimation accuracy within the seen distribution. We validated on a dataset of 94 built public projects registered in the Public Procurement Service (PPS) of the Republic of Korea, where BIM models were reconstructed from construction drawings at level of development (LOD) 200. Project-level input features are aggregated from element properties of walls, slabs, columns, apertures, stairs, and building storeys (counts and basic dimensions), are paired with total construction cost reported at PPS.

Rather than fixing transformation parameters a priori, the validation analyzes how the feature-wise reference scale τ and the in-distribution reference threshold r_{seen} affect the identification of unseen regions in the input feature space. Regions identified as unseen are then subject to shape regulation. Both τ and r_{seen} are parameterized as quantile levels of the training feature distributions, those varied over the set $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$. For each configuration, model behavior is evaluated using complementary metrics: mean absolute percentage error (MAPE) for finalized project states, and early-stage extrapolation metrics, namely the stepwise monotonicity violation rate (SMVR) and the zero-state error (ZSE).

MAPE is computed on finalized BIM states only, where predictions are expected to remain within the training distribution, to verify that the transformation does not degrade in-distribution accuracy. To evaluate early-stage extrapolation, simulated BIM authoring trajectories are generated from each final-state BIM model by randomly ordering the relevant elements and incrementally adding them, updating aggregated features at each step from the zero-state onward. ZSE evaluates the residual prediction magnitude at the zero-state, corresponding to the earliest mock-up state with no modeled elements, and is defined as

$$ZSE = \mathbb{E}(|\hat{y}(t)| \mid t = 0).$$

In contrast, SMVR evaluates extrapolation behavior along simulated authoring sequences. Let $\{x_s\}_{s=1}^S$ denote a sequence of BIM authoring states ordered by increasing authoring progress, such that quantity-type features monotonically increase between consecutive states, and let $\hat{y}(x_s)$ be the estimated cost at state x_s . SMVR is defined as:

$$SMVR = \frac{|\{s \mid \hat{y}(x_s) < \hat{y}(x_{s-1}), s = 2, \dots, S\}|}{S - 1}.$$

Lower SMVR indicate that estimates that evolves consistently with increasing modeled quantities, whereas higher values reflect frequent violations of positive monotonicity. In this study, SMVR is evaluated primarily in the low-count, out-of-distribution regime.

The estimators are implemented using an extreme gradient boosting regressor (XGBoost) (Chen and Guestrin, 2016), implemented with ‘xgboost.XGBRegressor’ Python library. Hyperparameters are optimized using the ‘Optuna’ library, resulting in a learning rate of 0.004, 1,800 estimators, a maximum tree depth of 15, a subsample ratio of 0.63, a column subsample ratio of 0.875, and a minimum child weight of 2.88. Train/validation/test splits are constructed using an 80/10/10 ratio. MAPE is reported on the held-out test set under finalized design conditions. By jointly analyzing MAPE and shape-related metrics across the (τ, r_{seen}) parameter grid, validation clarifies how aggressively early-stage predictions are anchored toward domain-consistent monotonic behavior while maintaining fidelity to patterns learned from finalized project data.

RESULTS

Figure 3 summarizes the effect of the proposed t-dependent origin anchoring transformation across the grid of τ and r_{seen} reference percentiles. Performance is evaluated using MAPE on finalized BIM states and SMVR measured along simulated early-stage BIM authoring trajectories.

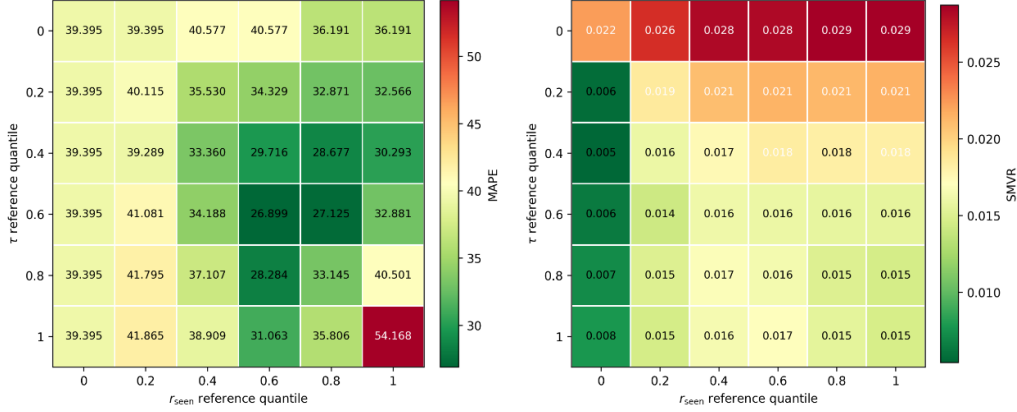


Figure 3. Effect of t-dependent origin anchoring on estimation accuracy and monotonicity across τ and r_{seen} reference percentiles.

Figure 3A reports MAPE values computed on finalized design states, where all input features lie within the training distribution. The baseline estimator without t-dependent anchoring yields a constant MAPE of 39.395 across all configurations. In contrast, applying the proposed transformation leads to substantial accuracy improvements for a wide range of τ and r_{seen} reference percentile combinations. In particular, intermediate reference percentiles result in markedly lower MAPE values, with the minimum MAPE of 26.899 observed when both τ and r_{seen} are defined at the 60th percentile of the training distributions. This indicates that partially treating weakly supported input regions as unseen can improve estimation accuracy by mitigating overconfident exploitation of learned patterns. Conversely, overly conservative reference percentiles lead to degraded performance, reflecting excessive suppression of valid in-distribution relationships.

Figure 3B reports SMVR values evaluated in the low-count, out-of-distribution regime. Without the proposed transformation, the baseline estimator exhibits substantial extrapolation instability, with an SMVR of 0.051 and a large ZSE (24.9 billion Korean Won $\approx$ 17.2 US Dollar), indicating both frequent monotonicity violations and severe non-zero predictions for near-empty states. Applying the t-dependent origin anchoring transformation consistently suppresses monotonicity violations across the entire τ and r_{seen} parameter grid, reducing SMVR to a narrow range of approximately 0.008-0.029. Compared to the baseline, this corresponds to a relative reduction of roughly 60%-84% in monotonicity violations. Importantly, parameter configurations that achieve the largest MAPE improvements also maintain low SMVR values, demonstrating that extrapolation robustness can be improved without sacrificing, and in some cases improving, in-distribution estimation accuracy. When the transformation is applied, ZSE is identically zero by construction, as predictions are explicitly anchored at the origin for empty states.

Taken together, the results demonstrate that the proposed t-dependent origin anchoring transformation enables a controllable inference-time trade-off between extrapolation robustness and adherence to learned in-distribution behavior. The parameter region defined by intermediate τ and r_{seen} values corresponds to a balanced exploration-exploitation regime, in which domain-

consistent behavior is enforced under sparse early-stage inputs while preserving, and in some cases improving, estimation accuracy on finalized design states.

CONCLUSION

Recent advances in AI are expanding early-stage design exploration, but cost feedback often arrives too late to support decision-making during BIM authoring. A key barrier is that early design states are underrepresented in the training data used for cost estimators. As a result, trained estimators often show extrapolation violations when applied to sparse early-stage BIM states.

To address this, we proposed a t -dependent origin anchoring transformation that imposes parameterized shape control at inference time without retraining the base estimator. The method computes a progress index $t(x) \in [0,1]$ from quantity-type features and modulates predictions to anchor near-empty states to zero and suppress monotonicity violations in weakly supported regions, while recovering the unconstrained estimator as inputs enter the seen regime. The two parameters, feature-wise reference scale τ and in-distribution reference threshold r_{seen} , provide an interpretable knob to control how aggressively early-stage inputs are treated as unseen.

Validation on 94 PPS public projects reconstructed at LOD 200 and evaluated on 94 simulated BIM authoring trajectories shows that extrapolation robustness can be improved without sacrificing in-distribution accuracy. The baseline estimator achieved a constant MAPE of 39.4% on finalized states, while applying the transformation reduced MAPE to 26.9% at optimal (τ, r_{seen}) settings. Along early-stage trajectories, SMVR decreased from 0.051 to 0.008–0.029 (60–84% reduction in monotonicity violations), and ZSE at the baseline estimator at 2.49×10^{10} KRW was eliminated with the transformation. Together, these results indicate that finalized-project datasets can support estimators that remain reliable when deployed during early-stage BIM authoring, enabling practical information front-loading for AI-augmented decision-making.

This study is limited to a single base cost estimator and simulated authoring trajectories derived from completed projects. Future work should validate the approach under live authoring traces and extend it extended project targets such as schedule, energy, and carbon metrics, including integration with uncertainty-aware estimation. More broadly, the results suggest that explicit extrapolation is critical for reliable early design phase decision-support, and that inference-time shape control provides a practical foundation for information front-loading in BIM workflows.

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